

Avinash Eranki, Ph.D.

Assistant Professor

Department of Biomedical Engineering
Indian Institute of Technology Hyderabad

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Work Experience

Assistant Professor, June 2020 – present

Department of Biomedical Engineering
Indian Institute of Technology Hyderabad
Principal Investigator of Medical Ultrasound Research Laboratory

Chief Executive Officer, October 2021 – present

Eranki Labs Private Limited, Hyderabad

Senior Research & Development Staff Engineer, May 2014 – March 2020

(R&D Staff Engineer III)
Sheikh Zayed Institute for Pediatric Surgical Innovation, Children's National Medical Center, Washington, DC

Research Collaborator, July 2014 – Present

Laboratory of Diagnostic Radiology Research, Radiology and Imaging Sciences Department, National Institutes of Health, Bethesda, MD

Special Research Volunteer, 2010 –2013

Functional & Applied Biomechanics Section, Rehabilitation Medicine Department
National Institutes of Health, Bethesda, MD

Researcher, 2011 –2012

Inova Regional Trauma Center (Level I), Inova Fairfax Hospital, Fairfax, VA

Graduate Research Assistant, 2007 –2013

Department of Electrical & Computer Engineering, George Mason University, Fairfax, VA

Education

Doctor of Philosophy, Electrical Engineering/Medical Imaging, Oct 2019

Thesis: Image-Guided High-Intensity Focused Ultrasound based Boiling Histotripsy for Treatment of Neuroblastoma
Utrecht University, Utrecht, Netherlands

Master of Science, Electrical Engineering, Jan 2011

Thesis: Measurement of Musculoskeletal Motion using Vector Tissue Doppler Imaging
George Mason University, Virginia, USA

Bachelor of Engineering, Electronics & Communication, May 2007

Visveswariah Technological University, Bangalore, India

Research

Extramural Research Grants (after joining IIT Hyderabad)

- Department of Science and Technology, 2025: Integrating Ultrasound Imaging with Therapeutic Ultrasound Device for Breast Tumor Therapy. **Role: Principal Investigator (₹9,998,000)**
- Indian Council for Medical Research – Intermediate Grant, 2024: Development of indigenous Robotic Ultrasound for synchronous management of Tumor motion and Radiation Hyperthermia. **Role: Principal Investigator (₹55,100,000)**
- Indian Council for Medical Research – Small Grant, 2024: Ultrasound-triggered Active Drug Delivery (uADD) System for Triple Negative Breast Cancer Therapy. **Role: Principal Investigator (₹5,900,000)**
- Department of Science and Technology, 2021: Noninvasive Diagnosis of Breast Cancer using Ultrasound-based Liquid Biopsy in a Point-of-Care Setting. **Role: Principal Investigator (₹3,576,758 – year 1)**
- BIRAC-Department of Science and Technology, 2021: Portable Ultrasound Device for Non-Invasive Therapy of Solid Tumors. **Role: Principal Investigator (₹5,000,000)**
- Science and Engineering Research Board, 2021: Dual-Mode Ultrasound for Soft Tissue Sarcoma Therapy. **Role: Principal Investigator (₹3,300,000)**
- Indian Council for Medical Research, 2021: Center of Excellence for Medical Devices. **Role: Co-Principal Investigator (₹157,100,000)**

Extramural Research Grants (before joining IIT Hyderabad)

- NIH National Cancer Institute, 2019: Bench to Bedside: Non-Invasive Treatment of Tumor in Children. Role: Associate Investigator (\$333,696)
- Board of Visitors Grant, Children's National Health System, 2017: Novel Non-Invasive Treatment of Pediatric Neuroblastoma. Role: Co-Principal Investigator (\$50,000)
- Joseph R Robert Jr. Endowment Award, 2015 & 2016: Immunogenic Effects of High Intensity Focused Ultrasound in a Metastatic Murine Tumor Model. Role: Co-Principal Investigator (\$25,000-year 1, \$50,000-year 2)

Awards

1. **Faculty Research Excellence Award** at IIT Hyderabad - 2024
2. 2020 '**Bracco Distinguished Young Investigator Award**' by the Focused Ultrasound Foundation, USA, for contribution to the field of Therapeutic Ultrasound. → **First Indian to receive this award**

3. **Best Start-up Presentation (Eranki Labs Private Limited)** at 2022 MedTech Symposium – Center for Healthcare Entrepreneurship at IIT Hyderabad

Clinical Trials (4 Clinical Trials Initiated after joining IIT Hyderabad)

1. Measuring Response to Neo-Adjuvant Chemotherapy for Breast Cancer using Novel Ultrasound Device and Quantitative Multi-Omics Data – **CTRI/2025/07/091416 → PI**

CTRI No	CTRI/2025/07/091416 [Registered on: 22/07/2025] Trial Registered Prospectively		
Acknowledgement Number	REF/2025/07/109297		
Last Modified On:	25/07/2025		
Post Graduate Thesis	No		
Type of Trial	Observational		
Type of Study	Cohort Study		
Study Design	Single Arm Study		
Public Title of Study	Ultrasound, Immunology and Genomics for Understanding Therapy Response in Breast Cancer		
Scientific Title of Study	Measuring the Response to Neo-Adjuvant Chemotherapy for Breast Cancer using Novel Ultrasound Device and Quantitative Multi-Omics Data - A pilot study		
Trial Acronym	NIL		
Secondary IDs if Any	Secondary ID		Identifier
	NIL		NIL
Details of Principal Investigator or overall Trial Coordinator (multi-center study)	Name	Dr S Annappurna	
	Designation	Associate Professor	
	Affiliation	AIIMS Bibinagar	
		Department of Radiodiagnosis, All India Institute of Medical Sciences Bibinagar, Hyderabad	
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Details Contact Person Scientific Query	Name	Dr Avinash Eranki	
	Designation	Assistant Professor	
	Affiliation	IIT Hyderabad	
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Details Contact Person Public Query	Name	Dr Avinash Eranki	
	Designation	Assistant Professor	
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2. Quantifying Breast Tumor texture and Microvasculature in 3D using Ultrasound Imaging – **CTRI/2024/07/069858 → PI**

CTRI No	CTRI/2024/07/069858 [Registered on: 03/07/2024] Trial Registered Prospectively		
Acknowledgement Number	REF/2024/07/087806		
Last Modified On:	25/07/2025		
Post Graduate Thesis	No		
Type of Trial	Observational		
Type of Study	Cohort Study		
Study Design	Single Arm Study		
Public Title of Study	3D Mapping of Breast Tumor Texture and Vasculature using Ultrasound Imaging		
Scientific Title of Study	Quantifying Breast Tumor Texture and Microvasculature in 3D using Ultrasound Imaging		
Trial Acronym	NIL		

3. Breast Tumor Heterogeneity Quantification using Ultrasound-based Shearwave Elastography – CTRI/2023/07/054888 → PI

CTRI No	CTRI/2023/07/054888 [Registered on: 06/07/2023] Trial Registered Prospectively	
Acknowledgement Number	REF/2023/06/068701	
Last Modified On:	01/07/2023	
Post Graduate Thesis	Yes	
Type of Trial	Observational	
Type of Study	Cross Sectional Study	
Study Design	Single Arm Study	
Public Title of Study	Breast tumor imaging using ultrasound	
Scientific Title of Study	Breast Tumor Heterogeneity Quantification using Ultrasound-based Shearwave Elastography	
Trial Acronym	NIL	
Secondary IDs if Any	Secondary ID NIL	
Details of Principal Investigator or overall Trial Coordinator (multi-center study)	Name	Avinash Eranki PhD
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	Affiliation	Indian Institute of Technology Hyderabad
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Details Contact Person Scientific Query Clarification(s) with Reply Modification(s)	Name	Avinash Eranki
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4. Comparing Genomic Variation Patterns in Blood and Tumor Tissue in Patients with Triple Negative Breast Cancer - CTRI/2022/02/040498 → PI

CTRI Number	CTRI/2022/02/040498 [Registered on: 22/02/2022] Trial Registered Prospectively	
Last Modified On:	18/02/2022	
Post Graduate Thesis	No	
Type of Trial	Observational	
Type of Study	Case Control Study	
Study Design	Other	
Public Title of Study	Comparing GENomic patterns in Triple negative breast cancer:	
Scientific Title of Study	Comparing Genomic Variation Patterns in Blood and Tumor Tissue in Patients with Triple Negative Breast Cancer	
Secondary IDs if Any	Secondary ID NIL	
Details of Principal Investigator or overall Trial Coordinator (multi-center study)	Name	Avinash Eranki Department of Biomedical Engineering Indian Institute of Technology, Hyderabad Sangareddy.
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	Name	Avinash Eranki Department of Biomedical Engineering Indian Institute of Technology, Hyderabad Sangareddy.
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Details Contact Person Public Query Modification(s)	Name	Avinash Eranki Department of Biomedical Engineering Indian Institute of Technology, Hyderabad Sangareddy.
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	Source of Monetary or Material Support	Department of Science and Technology, Govt of India
	Primary Sponsor	Name Department of Science and Technology Government of India Address Technology Bhavan, New Mehrauli Road, New Delhi 110016. Type of Sponsor Government funding agency

5. Treating neuroblastoma with Boiling Histotripsy and Nivolumab → **Co-I, First in the world at UMC Utrecht, the Netherlands**

Patents

1. **Eranki, Avinash.** (2021). Systems, Apparatuses and Methods for Focused Ultrasound Based Non-Invasive Biopsies. Indian Patent Application No. 202141056355. Indian Patent Office.
2. **Eranki, Avinash.** (2022). Systems, Apparatuses, and methods for dual mode activation in focused ultrasound procedures. Indian Patent Application No. 202241027757. Indian Patent Office.
3. **Eranki, Avinash** and R, Megha. (2022). Systems, Apparatuses and Methods for Activation State Control in Focused Ultrasound Based Procedures. Indian Patent Application No. 202241027756. Indian Patent Office.
4. **Eranki, Avinash** and Sabale, Aniket. (2023). Systems, Apparatuses and Methods for Imaging Control in Radiation Based Therapy Procedures. Indian Patent Application No. 202341020601. Indian Patent Office.

Publications

Journal Manuscripts

1. Sabale A, Rizvi MS, Chinthapenta V, **Eranki A.** Effect of focused ultrasound on shearwave production in a hyperelastic media. ***Biomechanics and Modeling in Mechanobiology.*** 2025 May 18:1-6.
2. Gopal TV, Ghori I, **Eranki A**, Hasan AQ, John R. Analysis of Doppler Audio Signals from the Carotid Artery. ***Journal of Medical and Biological Engineering.*** 2025 Apr;45(2):198-210.
3. Panda SS, Velladurai M. et al., CEUS and Quantitative Ultrasound Imaging Differentiate Malignant from Benign BIRADS 4 Breast Tumors: A Prospective Trial, ***European Radiology*** (under review) 2025.
4. Ramaviswanathan M, Sivaramalingam G, Radhakrishna S, **Eranki A**, "Sparse Volumetric Ultrasound Texture Analysis of Breast Fibroadenoma Tumors", ***Journal of Ultrasound in Medicine*** (under review) 2025.
5. TV VG, Ghori I, **Eranki A**, John R. MFCC-based Classification of Carotid Artery Doppler Audio Signals Using LSTM Network. ***Applied and Computational Engineering.*** 2024 Dec 19;100:36-42.
6. S. Chowdhury, S. J. A., **A. Eranki*** and S. K. Garlapati, Low-Cost Desktop Printed Sensors for Therapeutic Ultrasound Applications, ***IEEE Sensors Journal***, vol. 24, no. 23, pp. 39719-39726, 1 Dec.1, 2024, doi: 10.1109/JSEN.2024.3470223. ***Corresponding author**
7. Vigneshwaran G, Hasan QA, Kumar R, **Eranki A.** Analysis of single-nucleotide polymorphisms in genes associated with triple-negative breast cancer. ***Front Genet.*** 2022;13.
8. **Eranki A**, Srinivasan P, Ries M, Kim A, Lazarski CA, Rossi CT, Khokhlova TD, Wilson E, Knoblach SM, Sharma KV, Wood BJ, et al. High-Intensity Focused Ultrasound (HIFU)

Triggers Immune Sensitization of Refractory Murine Neuroblastoma to Checkpoint Inhibitor Therapy. **Clinical Cancer Research**. 2020 Mar 1;26(5):1152-61.

9. Spiliopoulos M*, Kuo CY*, **Eranki A***, Jacobs M, Rossi CT, Iqbal SN, Fisher JP, Fries MH, Kim PC. Characterizing placental stiffness using ultrasound shear-wave elastography in healthy and preeclamptic pregnancies. **Archives of Gynecology and Obstetrics**. 2020 Jul 16:1-0. * Equal contribution
10. Lau LW, **Eranki A**, Celik H, Kim A, Kim PC, Sharma KV, Yarmolenko PS. Are Current Technical Exclusion Criteria for Clinical Trials of Magnetic Resonance–Guided High-Intensity Focused Ultrasound Too Restrictive? Early Experiences at a Pediatric Hospital. **Journal of Ultrasound in Medicine**. 2020 Mar 30.
11. **Eranki A**, Mikhail AS, Negussie AH, Katti PS, Wood BJ, Partanen A. Tissue-mimicking thermochromic phantom for characterization of HIFU devices and applications. **International Journal of Hyperthermia**. 2019 Jan 1;36(1):517-28.
12. **Eranki A**, Farr N, Partanen A, Sharma KV, Rossi CT, Rosenberg AZ, Kim A, Oetgen M, Celik H, Woods D, Yarmolenko PS, et al. Mechanical fractionation of tissues using microsecond-long HIFU pulses on a clinical MR-HIFU system. **International Journal of Hyperthermia**. 2018 Nov 17;34(8):1213-24.
13. Yarmolenko PS, **Eranki A**, Partanen A, Celik H, Kim A, Oetgen M, Beskin V, Santos D, Patel J, Kim PC, Sharma K. Technical aspects of osteoid osteoma ablation in children using MR-guided high intensity focussed ultrasound. **International Journal of Hyperthermia**. 2018 Jan 2;34(1):49-58.
14. Sharma KV, Yarmolenko PS, **Eranki A**, Partanen A, Celik H, Kim A, Oetgen M, Kim PC. Magnetic resonance imaging–guided high-intensity focused ultrasound applications in pediatrics: early experience at children's national medical center. **Topics in Magnetic Resonance Imaging**. 2018 Feb 1;27(1):45-51.
15. **Eranki A**, Farr N, Partanen A, V. Sharma K, Chen H, Rossi CT, Kothapalli SV, Oetgen M, Kim A, H. Negussie A, Woods D, et al. Boiling histotripsy lesion characterization on a clinical magnetic resonance imaging-guided high intensity focused ultrasound system. **PloS one**. 2017 Mar 16;12(3):e0173867.
16. Sharma KV, Yarmolenko PS, Celik H, **Eranki A**, Partanen A, Smitthimedhin A, Kim A, Oetgen M, Santos D, Patel J, Kim P. Comparison of noninvasive high-intensity focused ultrasound with radiofrequency ablation of osteoid osteoma. **The Journal of pediatrics**. 2017 Nov 1;190:222-8.
17. Kuo CY, **Eranki A**, Placone JK, Rhodes KR, Aranda-Espinoza H, Fernandes R, Fisher JP, Kim PC. Development of a 3D printed, bioengineered placenta model to evaluate the role of trophoblast migration in preeclampsia. **ACS Biomaterials Science & Engineering**. 2016 May 6;2(10):1817-26.
18. **Eranki A**, Cortes N, Ferenčec ZG, Sikdar S. A novel application of musculoskeletal ultrasound imaging. **JoVE (Journal of Visualized Experiments)**. 2013 Sep 17(79):e50595.
19. Lebedowska MK, Sikdar S, **Eranki A**, Garmirian L. Knee joint angular velocities and accelerations during the patellar tendon jerk. **Journal of neuroscience methods**. 2011 Jun 15;198(2):255-9.

Conference Publications

1. **Eranki A** and S. Sikdar, "Experimental characterization of a vector Doppler system based on a clinical ultrasound scanner," *Proc. the 31th Annual International Conference of IEEE EMBS*, pp. 2260-2263, 2009.
2. S. Sikdar, M. Lebedowska, **Eranki A**, L. Garmirian, and D. Damiano, "Measurement of rectus femoris muscle velocities during patellar tendon jerk using vector tissue Doppler imaging," *Proc. the 31th Annual International Conference of IEEE EMBS*, pp. 2963-2966, 2009.
3. **Eranki A**, L. Bellini, L. Prosser, C. Stanley, D. Bland, K. Alter, D. Damiano and S. Sikdar, "Measurement of tendon velocities using vector Tissue Doppler Imaging: A feasibility study," *Proc. the 32th Annual International Conference of IEEE EMBS*, pp. 5310-5313, 2010.
4. **Eranki A**, K. Almuhanha and S. Sikdar, "Characterization of a Vector Doppler System based on an Array Transducer", *Proc. the 2010 IEEE International Ultrasonics Symposium*, pp: 1076 – 1079, 2010.
5. **Eranki A**, S. Sikdar, C. Stanley, L. Prosser, L. Curatalo, D. Bland, K. Alter and D. Damiano, "Measuring Tendon Velocities using Vector Tissue Doppler Imaging", *Proc. the 160th Meeting of the Acoustical Society of America*, pp.2305-2305, 2010.
6. P. Otto, L. Curatalo, **Eranki A**, L. Prosser, D. Damiano, K. Alter and S. Sikdar, "Quantitative Tracking of Tendon Motion From Ultrasonic Imagery using Curved M-Mode", *Proc. the 36th International Symposium on Ultrasonic Imaging and Tissue Characterization*, 2011.
7. **Eranki A**, P. Otto, L. Curatalo, L. Prosser, K. Alter, D. Damiano and S. Sikdar, "Measurement of Tendon Velocities using Vector Tissue Doppler Imaging and Curved M-Mode in Patients with Cerebral Palsy", *Proc. the 2011 IEEE International Ultrasonics Symposium*, pp. 676 – 679, 2011.
8. K. N. Mulpuri, N. S. Hussein, **Eranki A** and S. Sikdar, "Understanding Dynamics of Oxygenation and Blood Flow in the Upper Trapezius Muscle During Exercise," *Proc. the Annual International Conference of BMES*, 2011.
9. **Eranki A**, N. Cortes, Z.G. Ferenček, J.J. Kim and S. Sikdar, "Real-Time Measurement of Rectus Femoris Muscle Kinematics during Drop Jump using Ultrasound Imaging: A Preliminary Study," *Proc. the 34th Annual International Conference of IEEE EMBS*, pp. 4851-4854, 2012.
10. M. Zaazhoa, **Eranki A**, L. Gerber, K. Armstrong and S. Sikdar, "Oxygen Kinetics in patients with Active Myofascial Trigger Points in Upper Trapezius Muscle using near Infrared Spectroscopy," *Proc. the Annual International Conference of BMES*, 2012.
11. B. Dougherty, **Eranki A**, L. Curatalo, D. Damiano, and S. Sikdar, "Ultrasound Echotexture Analysis of the Tibialis Anterior Muscle during Isometric Contraction in Children with Cerebral Palsy and Healthy Controls," *Proc. the Annual International Conference of BMES*, 2012.
12. **Eranki A**, N. Cortes, Z.G. Ferenček and S. Sikdar, "Dynamic Musculoskeletal Imaging using a USB-based Ultrasound Probe for Point-of-Care Applications," *Proc. the Annual International Conference of IEEE EMBS Point-of-Care Health Technologies*, pp. 18-21, 2013.
13. M. Zaazhoa, **Eranki A**, L. Gerber and S. Sikdar, "Comparison of Oxygen Kinetics in the Upper Trapezius in Patients with Chronic Neck Pain and Myofascial Trigger Points Before and After Treatment," *Proc. the Annual International Conference of BMES*, 2013.
14. A.J.Aralar, **Eranki A**, D. Turo, C. Stanley, D. Damiano, S. Sikdar, "Ultrasound Characterization Of The Tibialis Anterior Muscle In Children With Cerebral Palsy And Foot Drop," *Proc. the Annual International Conference of BMES*, 2014.
15. D. Yang, H.Celik, D.Wackerle, D.Kinnaird, **Eranki A**, M.Oetgen, A.Kim, K.Sharma, H.Kim, P.Kim, P.S. Yarmolenko, "Optical Measurement of Skin Temperature in MR-HIFU," *Proc. the Annual International Conference of FUSF*, 2014.
16. D.Kinnaird, D.Wackerle, D.Yang, M.Oetgen, **Eranki A**, A.Kim, K.Sharma, H.Kim, P.Kim, P.S. Yarmolenko, H.Celik, "Treatment planning and patient positioning for MR-guided high intensity focused ultrasound treatment: a systematic approach," *Proc. the Annual International Conference of FUSF*, 2014.

17. D.Wackerle, H.Celik D.Kinnaird, D.Yang, M.Oetgen, **Eranki A**, A.Kim, K.Sharma, P.Kim, P.S. Yarmolenko," The Optimization of Treatment Planning and Ablation Rate Improvements on Feasibility of Pediatric MR-HIFU Applications," *Proc. the Annual International Conference of FUSF*, 2014.
18. C.Kuo, **Eranki A**, J.K.Placone, K.R.Rhodes, P.C.Kim, R. Fernandes, J.P.Fisher, "Development of a 3D-Printed Tissue Engineered Invasion Model of Placental Biology," *Proc. TERMIS world Congress*, 2015.
19. **Eranki A**, Navid Farr, Ari Partanen, Pavel S. Yarmolenko, Ayele H. Negussie, Karun V. Sharma, Haydar Celik, Bradford J. Wood and Peter C. Kim, "Volumetric and Thermal Characterization of Boiling Histotripsy Lesions Based on a Clinical MR-HIFU System," *Proc. International Society for therapeutic Ultrasound*, 2016.
20. **Eranki A**, N. Farr, A. Partanen, K. V. Sharma, P.S. Yarmolenko, B. J. Wood P. C. W. Kim," Quantification of Cavitation Histotripsy Lesion Volumes on a Clinical MR-HIFU System," *Proc. the Annual International Conference of FUSF*, 2016.
21. N. Farr, M. Alnazeer, P. Yarmolenko, P. Katti, A. Partanen, **Eranki A**, P. C.W. Kim, B. J. Wood," Retrospective study of patient suitability for MR-HIFU liver therapy," *Proc. International Society for therapeutic Ultrasound*, 2016.
22. N. Farr, A.Partanen, S.V.V.N. Kothapalli, **Eranki A**, A.Negussie, E. Wilson, R. Seifabadi, P. Kim, H. Chen, B. J. Wood," Tissue-mimicking thermochromic rib phantom for characterization of intercostal HIFU sonications and beam-shaping methods," *Proc. International Society for therapeutic Ultrasound*, 2016.
23. K.V. Sharma, P.S.Yarmolenko, A.Partanen, H. Celik, **Eranki A**, V. Beskin, D. Santos, J. Patel, M. Oetgen, A. Kim, P. C. W. Kim, "Towards NonInvasive Treatment of Osteoid Osteoma: State of the Art in Context of Accepted Methods," *Society for Thermal Medicine*, 2016.
24. P.S. Yarmolenko, H. Celik, **Eranki A**, P. C.W. Kim, K. V. Sharma, "Feasibility, safety, and tolerability of osteoid osteoma ablation in children using the Philips Sonalleve V2 MR-HIFU system," *Proc. International Society for therapeutic Ultrasound*, 2016.
25. K. Sharma, P. Yarmolenko, H. Celik, **Eranki A**, A. Kim, M. Oetgen, A. Partanen, A. Smitthimedhin, J. Patel, P. C.W. Kim," Changing paradigm for treatment of osteoid osteoma in children," *Journal of Vascular and Interventional Radiology*, Volume 28, Issue 2, Page S94, 2017.
26. Spiliopoulos M, Kuo CY, **Eranki A**, Iqbal S, Fisher JP, Fries M, Kim PC. 467: Determining In-vivo placental stiffness in healthy and preeclamptic pregnancies using shear-wave elastography. *American Journal of Obstetrics & Gynecology*. 2018 Jan 1;218(1):S281.
27. Kim A, Sharma K, Yarmolenko P, Geller J, Pressey J, Racadio J, Celik H, **Eranki A**, Lanier M, Tydings C, Kaplan R, Partanen A, and P.C.W.Kim. Safety and feasibility of magnetic resonance-guided high intensity focused ultrasound (MR-HIFU) for the ablation of relapsed or refractory pediatric solid tumors including desmoid tumors. *Annual International Meeting of Connective Tissue Oncology Society*. 2019
28. Tydings C, Yarmolenko P, Geller J, Pressey J, Ricardio J, Celik H, **Eranki A**, Lainier M, Sharma K, and Kim A. Desmoid Tumor Volumetric Analysis for Disease Response post Ablation with Magnetic Resonance Guided High Intensity Focused Ultrasound (MR-HIFU). *Annual International Meeting of Connective Tissue Oncology Society*. 2019
29. **Eranki A**, Ries A, Srinivasan P, Sandler A, Kim A, Sharma KV, Kim PCW, Wood BJ, Moonen CTW. Boiling Histotripsy Increases Systemic Immune Response against Murine Neuroblastoma Tumor and leads to Long-Term Survival. *European Society for Molecular Imaging: TOPIM 2020 on Imaging Immunity*. 2019
30. **Eranki A**, Ries A, Srinivasan P, Wu X, Khokhlova TD, Kim A, Sharma KV, Sandler AD, Kim PCW, Wood BJ, Moonen CTW. Immune sensitization and therapeutic impact of boiling histotripsy in refractory murine neuroblastoma. *Focused Ultrasound Symposium*, 2020.
31. **Eranki A**, Ries A, Srinivasan P, Wu X, Khokhlova TD, Kim A, Sharma KV, Sandler AD, Kim PCW, Wood BJ, Moonen CTW. Temporal Dynamics of Intratumoral Immune Cell Infiltration Triggered by Boiling Histotripsy. *Focused Ultrasound Symposium*, 2020.

32. Tydings C, **Eranki A**, Sharma KV, Kim A. High intensity focused ultrasound thermal ablation in combination with checkpoint inhibitors for the treatment of refractory murine neuroblastoma. Focused Ultrasound Symposium, 2020.
33. Tydings C, **Eranki A**, Sharma KV, Kim A. Immune effects of high intensity focused ultrasound ablation with checkpoint inhibitors in murine neuroblastoma. American Association for Cancer Research Annual Meeting, 2021.
34. Vigneshwaran Gurunadan, Hasan Q, **Eranki A**. 6P vAO - Variant Annotator for OncoKB: A simplified interface to identify targeted therapies for somatic tumors. ESMO Open [Internet]. Elsevier; 2025;10.
35. S. S. Panda, D. J. Mech, **Eranki A** and M. S. Rizvi, "Characterizing Ultrasound-based Shearwave Propagation in Collagen Fiber Network," 2024 IEEE South Asian Ultrasonics Symposium (SAUS), Gujarat, India, 2024, pp. 1-4, doi:10.1109/SAUS61785.2024.10563612.
36. R. Megha, Geethapriya, S. Radhakrishna and **Eranki A**, "Quantifying Textural Heterogeneity of Ultrasound Images in Breast Fibroadenomas," 2024 IEEE South Asian Ultrasonics Symposium (SAUS), Gujarat, India, 2024, pp. 1-4, doi:10.1109/SAUS61785.2024.10563460.
37. R. Megha, Geethapriya, S. Radhakrishna and **Eranki A**, "Breast Tumor Heterogeneity Quantification using 3D Ultrasound Texture," 2024 IEEE South Asian Ultrasonics Symposium (SAUS), Gujarat, India, 2024, pp. 1-4, doi:10.1109/SAUS61785.2024.10563639.
38. Sabale A, Rizvi MS, Chinthapenta V, **Eranki A**. Ultrasound wave propagation and shear stress through tissues for intercostal focused ultrasound therapy. 2024 IEEE South Asian Ultrasonics Symposium (SAUS) 2024 Gujarat, India, (pp. 1-4). IEEE. 10.1109/SAUS61785.2024.10563729
39. Sabale A, Rizvi MS, Chinthapenta V, **Eranki A**. Effect of focused ultrasound on shear wave propagation in nonlinear media. In Structural Integrity Conference and Exhibition 2022 Dec 14 (pp. 219-228). Singapore: Springer Nature Singapore.
40. TV VG, Ghorl I, **Eranki A**, John R. Haemodynamic Parameters Analysis through Audio Signal Processing Analysis of Doppler Audio Signals for Carotid Artery Assessment. In Proceedings of the 11th International Conference on Bioinformatics Research and Applications 2024 Sep 13 (pp. 106-112).
41. TV VG, Ghorl I, **Eranki A**, John R. Early screening of cerebrovascular diseases through Doppler ultrasound analysis of carotid arterial blood flow. In 2024 international conference on signal processing and communications (SPCOM) 2024 Jul 1 (pp. 1-5). IEEE.

Book Chapters

1. Mech DJ, Murugappan S, Buddhiraju HS, **Eranki A**, Rengan AK, Rizvi MS. AI on DDS for regenerative. *Artificial Intelligence in Tissue and Organ Regeneration*. 2023 Aug 18:133.

Teaching and Training

Courses Taught

Year	Session	Course Code	Course	Number of Credits	Feedback Score
2021	Spring	BM5170	Ultrasound in Medicine	3	4.81
2022	Spring	BM2003	Introduction to Embedded Systems	3	4.38
		BM5170	Ultrasound in Medicine	3	5.00
		BM6006	Industry Lectures	1	NA

	Fall	BM6006	Industry Lectures	1	NA
2023	Spring	BM5170	Ultrasound in Medicine	3	4.00
		BM2003	Introduction to Embedded Systems	3	4.74
	Fall	BM4080	Medical Image Processing and Analysis	2	4.02
		BM4021	Medical Image Processing Lab	2	4.13
2024	Spring	BM5170	Ultrasound in Medicine	3	4.80
		BM2003	Introduction to Embedded Systems	3	4.72
	Fall	BM4080	Medical Image Processing and Analysis	2	4.06
		BM4021	Medical Image Processing Lab	2	3.98
2025	Spring	BM5170	Ultrasound in Medicine	3	5.00
		BM2003	Introduction to Embedded Systems	3	3.52
		BM3090	Biomedical Imaging	1	3.69
		BM6090	Biomedical Imaging	1	4.40
	Fall	BM4080	Medical Image Processing and Analysis	2	
		BM4021	Medical Image Processing Lab	2	
Total Number of Credits/Year (excluding 2021) : 10					

Mentoring Students

Doctoral Students

Year	Degree Program	Name	Status
2020	Ph.D. in BME	Aniket Arun Sabale	Graduated (PhD Defense Completed - 2025)
2021		Vigneshwaran G	Final Year
		Megha R	Final Year
2022		Shradha Suman Panda	4 th Year
2023		Shaswata Chowdhury	2 nd Year
2025		Uthra T K	1 st Year
2025		Charayyswerrya N	1 st Year

Master's Students

Year	Degree Program	Name	Status
2020	MTech in BME	Vivek P	Graduated
2021		G.Prithvi Shankar	Graduated
		Shaswata Chowdhury	Graduated
2024	MTech in MDI	Jyothika Kambhampati	On-going
		Md Tagidul Islam	On-going

Undergraduate Students (mentored for > 1 year post-BTech)

Year	Degree Program	Name	Status
2024	BTech in BME	Nitin Shreyas	Graduated (M.S. at UC San Diego)

Post-Doctoral Fellows

Year	Degree Program	Name	Status
2025	Ph.D.	Manish Chandra Joshi	Currently working with me

Professional Service

Peer Review and Editorial Roles

1. **Editorial Board Member** for Ultrasound in Medicine and Biology Journal- 2018 to date.
2. **Reviewer** for Ultrasound in Medicine and Biology, Ultrasonics, IEEE SAUS conference.
3. **Guest Editor** at Frontiers in Bioengineering and Biotechnology Journal (Q1).
4. **Reviewer** for Wadhwani Foundation, SRISTI GYTI grants, and Doctoral Thesis (IITGn).
5. **Member of the Board of Studies**, Nitte Meenakshi Institute of Technology, Bangalore, Karnataka, 2021 -2022.

Professional Societies & Universities

1. **Guest Researcher/Employee**, Division of Imaging and Oncology, University Medical Center Utrecht, Netherlands – Jan 2021 to Dec 2024.
2. **Expert member of Medical Devices Committee**, Bureau of Indian Standards, July 2021- to date.
3. **Active Member**, American Association for Cancer Research – Jan 2021 – to date.
4. **Member**, IEEE & UFFC Society, 2022 – to date.
5. Invited to attend & present the **5th workshop on Focused Ultrasound and Cancer Immunotherapy** (2023) held in Arlington, VA (only researcher from India).

Conference Organization

1. Helped organize and **chair one technical session** on the inaugural IEEE South Asian Ultrasonics Symposium – 2024.
2. **Conference Co-Chair** for the IEEE South Asian Ultrasonics Symposium – 2026.

Translational Effort and Collaborations

1. Created **Eranki Labs** Private Limited in October 2021 – Director and CEO.
 - a. Eranki Labs Private Limited designed, developed, and manufactured **India's First** High-Intensity Focused Ultrasound System in 2024.
 - b. Our HIFU device can automatically perform HIFU therapy with minimal human intervention. The device includes a robotic arm, which will house the HIFU transducer, the imaging ultrasound (US) transducers, and necessary electronics paired with a graphical user interface, which acts as a unified controlling system.

- c. Driven by an intelligent, 'hands-free' therapy algorithm, an ultrasound imaging-guided HIFU therapy portable device is developed to provide a non-invasive therapy of solid tumors, including breast tumors.
- d. This system is currently portable, cost-effective, and easy to use at primary and tertiary medical centers across India and elsewhere. This innovative and disruptive device will be the first in India to perform a wholly noninvasive and nonionizing radiation-based therapy for solid tumors under ultrasound imaging guidance.
- e. We have currently raised over 3.2 crores of funding and are in line to receive another 8-10 crores over the next two years.
- f. We are initiating India's first clinical trial involving an HIFU system to treat breast tumors.
- g. Our device is recognized by ICMR, BIRAC, and the Focused Ultrasound Foundation (the USA).

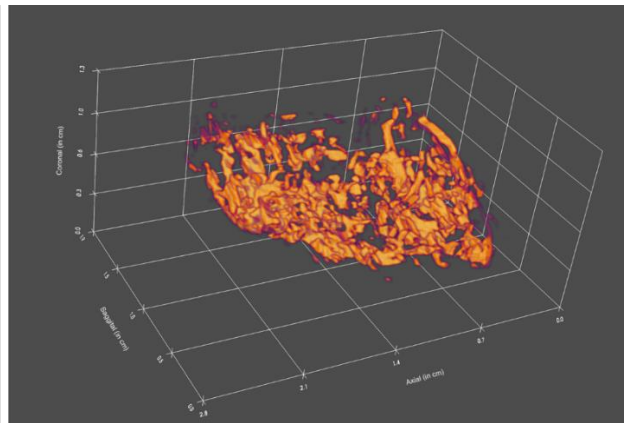
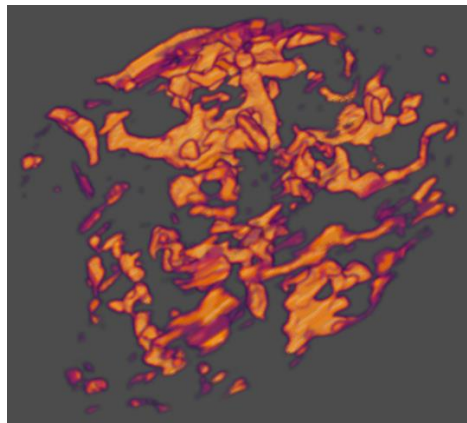


- h. Eranki Labs also created a **novel ultrasound scanner** that is able to build 3D volumes of tumor microvasculature, shearwave elastography (SWE), and b-mode. This scanner can be used on any ultrasound system and help build volumetric data from any ultrasound imaging mode.



Figure shows device to collect volumetric ultrasound images

- i. Using our novel device, we have mapped the microvasculature, morphology, and stiffness of over 200 breast tumors from patients *in vivo* using ultrasound imaging without the need for contrast agents. Figure below shows mapping breast intratumor microvasculature in 3D.



2. Created a company with three other co-founders in October 2025 – Co-Director and Co-Founder.
 - a. The company works on developing a predictive and prognostic suite for breast cancer diagnosis and therapy.
 - b. This product will be used by radiologists, oncologists, surgeons, and others for helping tailoring and personalizing therapies for women suffering from breast cancer.
3. Established a strong collaboration with **AIIMS Bibinagar, Amrita Hospital Faridabad, and Chennai Breast Center** to collect **India's First** prospective volumetric ultrasound database from breast tumors in patients.

4. Working with **Kamineni Institute of Medical Sciences and AIG Hospitals** to understand the breast cancer genomics and develop novel AI in pathology for hepatic steatosis.

Administration

1. Faculty-in-Charge Student Activities – 2021 to 2023
2. National Innovation and Startup Policy Committee Member – 2021 to 2023
3. Institute Innovation Council Member – 2021 to 2023
4. Food Court Committee Member – 2021 to 2023
5. BUILD Project Monitoring Committee Member – 2021 to 2023
6. Student Welfare Committee Member – 2021 to 2023
7. DPGC for Center for Interdisciplinary Programs (CIP) – 2024 to date
8. Faculty-in-Charge Hospital – 2023 to 2024
9. Academic Review Committee Member – 2025 to date
10. Research Advisory Council Member – 2024 to date
11. Space Committee Member – 2025 to date
12. Developed and implemented an MSc in Medical Physics Program along with other colleagues
13. Help build the Center of Excellence in Medical Devices funded by ICMR-DHR

Outreach

1. Delivered several lectures for scientists and students at DBT-NIAB, University of Hyderabad, and degree college teachers on multiple occasions.
2. Developed experiments to demonstrate sound and wave propagation for high school students.
3. Invited to give a keynote lecture at the IEEE International Conference on Robotics and Mechatronics (ICRM 2025) at Amritapuri Campus Kollam, Kerala.
4. Invited as a Speaker at the Translation of Bioengineering Research for Healthcare at NIT Rourkela – 11th – 13th December 2025.

Infrastructure Established (except first, rest are procured via Extramural Funding)

Item name	Quantity	Make
Ultrasound machine	1	Supersonic
HIFU Systems	2	Eranki Labs
Linear Stages - LBAS08-5-1050,LBAS08-5-500,LBAS08-5-250	3	Yamaha
Brightfield+Fluroscence Microscope	1	Nikon
Workstations	14	Lenovo/HP/AntPC
Collaborative Robots (6DOF, 5DOF, 4DOF)	5	IGUS/Systemantics
RF Power Amplifier	4	E & I
Digital Oscilloscope	1	Tektronix
Function Generator	1	Tektronix
RAID Server	1	Qsync
CO2 Incubator	1	ThermoFisher
Wireless Ultrasound machine	2	Guturu pvt ltd
HIFU Transducer	3	Sonic Concepts Inc
Ultrasound Transducers	4	Precision Acoustics
QuantStudio RT-PCR - 5 channel	1	ThermoFisher
Capsule Hydrophone	1	Onda Corp
Hydrophone Pre-Amplifier	1	Onda Corp
Semi Automatic Cryostat Microtome	1	WESWOX
Fully Automatic Cryostat Microtome	1	WESWOX
3M-Membrane	1	Liquicell
KNF pump(Laboport + small circ)	2	KNF
KNF pump DC 12V (Big circ)	1	KNF
Hot Air Oven for Sensors	1	MH Enterprises
Verical laminar air flow	1	Sterile Tech india
Temperature Controlled Centrifuge	1	MH Enterprises
Galil Controller - 4 axes	1	Galil Motion
Creality Ender 5 S1 3D Printer	1	WOL 3D India limited
Hydra 300 3D Printer	1	3ding
Robotic Controllers	5	IGUS(INDIA) PVT.LTD.
Dissolved Oxygen sensor	1	DFRobot
Open Architechture Ultrasound System	1	Sono Light
Refrigerator	1	Samsung
Control panel 200w motor with drive,cables	1	Strategi
Miscellaneous Items, RAM's, UPS's, Cameras, Motors, Furniture, etc.	>10	APC, Portescap, etc.